

Finite-Difference Formulation of a Gas Pulse-Decay Experiment

Derivation from mass conservation to an explicit numerical scheme

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Purpose and document status

These notes organize the ideal-gas, Darcy–Klinkenberg pulse-decay model and its explicit finite-difference discretization from first principles. Their immediate purpose is to serve as the mathematical specification for a small, transparent Python implementation. A later volume will repeat the derivation with a cell-centred finite-volume method and compare the two formulations.

Two boundary-node discretizations are retained deliberately:

1. the *direct dynamic-boundary update* derived by applying forward Euler in time and a one-sided pressure-gradient approximation to the tank boundary condition; and
2. the *storage-consistent boundary update* reported by [Jannot et al. \(2007\)](#), which assigns one half of the adjacent core interval’s pore storage to each boundary node.

The distinction matters at finite grid spacing. The first version is useful pedagogically and converges as the grid is refined, but it does not represent the two boundary half-intervals in its discrete storage. The second version is the preferred reference formulation for the first implementation.

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1. Problem definition

1.1. Physical configuration

The experiment consists of a cylindrical porous core of length L and cross-sectional area S , connected to finite upstream and downstream gas volumes V_0 and V_1 . At the beginning of an idealized pulse-decay test, the upstream tank is at the higher absolute pressure $P_{u,i}$, whereas the core and downstream tank are at the lower pressure $P_{d,i}$. Opening the valve creates transient gas flow in the positive x -direction.

The core-face pressures are assumed continuous with their well-mixed tank pressures:

$$P(0, t) = P_u(t), \quad P(L, t) = P_d(t). \quad (1.1)$$

The measured pressure traces are therefore the two dynamic boundary values of the core problem, not externally prescribed Dirichlet data.

1.2. Scope and assumptions

The first implementation uses the smallest model that retains gas compressibility and Klinkenberg slip. Its assumptions are:

1. one-dimensional axial flow through a homogeneous, isotropic, rigid core;
2. constant core geometry, porosity ε , intrinsic permeability k_ℓ , gas viscosity μ , and Klinkenberg coefficient b ;
3. single-phase, isothermal, ideal-gas behaviour;
4. Darcy flow at negligible Reynolds number, with no Forchheimer inertia and no gravity term;
5. well-mixed upstream and downstream tanks, so each tank has one spatially uniform pressure;
6. no external leakage, adsorption, reaction, or source term;

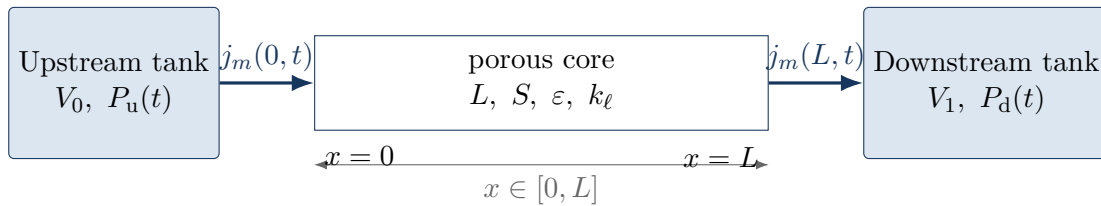


Figure 1.1.: Idealized pulse-decay configuration. Positive mass flux is defined from the upstream tank toward the downstream tank.

7. pressure and temperature equilibration in valves and connecting lines is not resolved explicitly; calibrated dead volumes are folded into V_0 and V_1 ;
8. all pressures are *absolute*.

These assumptions reproduce the classical complete ideal-gas Darcy–Klinkenberg initial-boundary-value problem used by Jannot et al. (2007) and revisited by Lasseux et al. (2012). ~~They are modelling choices rather than universal truths.~~ Real-gas density, variable viscosity, thermal transients, leakage, and uncertain initial expansion can be introduced only after this baseline problem has been verified.

1.3. Nomenclature

Symbol	Meaning	SI unit
b	Klinkenberg pressure coefficient	Pa
C_g	ideal-gas density–pressure coefficient, $M/(RT)$	$\text{kg m}^{-3} \text{Pa}^{-1}$
j_m	gas mass flux per unit gross area, positive toward $+x$	$\text{kg m}^{-2} \text{s}^{-1}$
k_a	pressure-dependent apparent gas permeability	m^2
k_ℓ	intrinsic (liquid-equivalent) permeability	m^2
L	core length	m
M	gas molar mass	kg mol^{-1}
N	number of equal spatial intervals	1
P	absolute gas pressure	Pa
P_u, P_d	upstream and downstream tank pressures	Pa
$P_{u,i}, P_{d,i}$	initial upstream and downstream pressures	Pa
R	universal gas constant	$\text{J mol}^{-1} \text{K}^{-1}$
S	core cross-sectional area, $S = \pi D^2/4$	m^2
t	time	s
T	uniform absolute temperature	K
u	superficial Darcy velocity, positive toward $+x$	m s^{-1}
V_0, V_1	effective upstream and downstream tank volumes	m^3
x	axial coordinate	m
$\Delta x, \Delta t$	spatial and temporal increments	m, s
ε	connected porosity	1
μ	dynamic gas viscosity	Pa s
ρ	gas mass density	kg m^{-3}
φ	transformed pressure, $(P + b)^2$	Pa^2

Notation choice. The source papers use e for core length, V_0 and V_1 for the two tank volumes, and ϕ for the transformed pressure. These notes use L , retain V_0, V_1 , and typeset the transformed pressure as φ . The physics is unchanged.

2. Governing equations

2.1. Ideal-gas density

At constant temperature, the ideal-gas law gives

$$\rho(P) = \frac{M}{RT}P = C_g P, \quad C_g \equiv \frac{M}{RT}. \quad (2.1)$$

Consequently,

$$\frac{\partial \rho}{\partial t} = C_g \frac{\partial P}{\partial t}. \quad (2.2)$$

The constant C_g will cancel from the ideal-gas governing equations, but retaining it during the derivation makes the mass balance explicit.

2.2. Darcy–Klinkenberg mass flux

The Klinkenberg relation expresses the apparent gas permeability as

$$k_a(P) = k_\ell \left(1 + \frac{b}{P}\right). \quad (2.3)$$

Darcy's law for the superficial velocity is

$$u = -\frac{k_a(P)}{\mu} \frac{\partial P}{\partial x}. \quad (2.4)$$

The mass flux per unit gross area is $j_m = \rho u$. Substituting equations (2.1) and (2.3) into equation (2.4) therefore gives

$$\begin{aligned} j_m &= -C_g P \frac{k_\ell}{\mu} \left(1 + \frac{b}{P}\right) \frac{\partial P}{\partial x} \\ &= -C_g \frac{k_\ell}{\mu} (P + b) \frac{\partial P}{\partial x}. \end{aligned} \quad (2.5)$$

For the usual pulse-decay ordering $P_u > P_d$, the pressure gradient is negative. Equation (2.5) then gives $j_m > 0$, consistent with the positive left-to-right flux convention in figure 1.1.

2.3. Core mass conservation

Local gas-mass conservation in the rigid porous core is

$$\frac{\partial(\varepsilon\rho)}{\partial t} + \frac{\partial j_m}{\partial x} = 0. \quad (2.6)$$

For constant porosity, substitution of equations (2.2) and (2.5) yields

$$\varepsilon C_g \frac{\partial P}{\partial t} - C_g \frac{k_\ell}{\mu} \frac{\partial}{\partial x} \left[(P + b) \frac{\partial P}{\partial x} \right] = 0. \quad (2.7)$$

Cancelling C_g gives the conservative pressure equation

$$\boxed{\varepsilon \frac{\partial P}{\partial t} = \frac{k_\ell}{\mu} \frac{\partial}{\partial x} \left[(P + b) \frac{\partial P}{\partial x} \right]}, \quad 0 < x < L, \quad t > 0. \quad (2.8)$$

Applying the product rule exposes the nonlinearity:

$$\boxed{\varepsilon \frac{\partial P}{\partial t} = \frac{k_\ell}{\mu} \left[(P + b) \frac{\partial^2 P}{\partial x^2} + \left(\frac{\partial P}{\partial x} \right)^2 \right]}. \quad (2.9)$$

The factor multiplying the spatial operator is k_ℓ/μ , not its reciprocal. This placement is required both by Darcy's law and by dimensional consistency.

2.4. Initial condition

The idealized pressure pulse is represented by

$$P(0, 0) = P_{u,i}, \quad (2.10a)$$

$$P(x, 0) = P_{d,i}, \quad 0 < x \leq L. \quad (2.10b)$$

Thus the core and downstream tank initially share the lower equilibrium pressure, while the upstream boundary starts at the pulse pressure. The initial discontinuity at $x = 0$ is an idealization of rapid valve opening. Measured post-expansion pressures should replace nominal pre-opening pressures when the experimental expansion is appreciable.

2.5. Dynamic tank boundary conditions

The boundary conditions follow from the mass balance of each finite tank. They are derived here with signs shown explicitly.

2.5.1. Upstream tank

The mass stored upstream is

$$m_0(t) = V_0 \rho(P_u(t)) = C_g V_0 P_u(t). \quad (2.11)$$

Positive $j_m(0, t)$ leaves this tank, so

$$\frac{dm_0}{dt} = -S j_m(0, t). \quad (2.12)$$

Using equations (1.1) and (2.5) gives

$$C_g V_0 \frac{\partial P}{\partial t}(0, t) = C_g \frac{k_\ell S}{\mu} [P(0, t) + b] \frac{\partial P}{\partial x}(0, t). \quad (2.13)$$

After cancellation,

$$\boxed{\frac{\partial P}{\partial t}(0, t) = \frac{k_\ell S}{\mu V_0} [P(0, t) + b] \frac{\partial P}{\partial x}(0, t)}. \quad (2.14)$$

During ordinary decay, $\partial P / \partial x < 0$, so $\partial P(0, t) / \partial t < 0$, as required.

2.5.2. Downstream tank

The downstream mass is

$$m_1(t) = C_g V_1 P_d(t). \quad (2.15)$$

Positive $j_m(L, t)$ enters this tank, so

$$\frac{dm_1}{dt} = +S j_m(L, t). \quad (2.16)$$

Substitution now gives

$$\boxed{\frac{\partial P}{\partial t}(L, t) = -\frac{k_\ell S}{\mu V_1} [P(L, t) + b] \frac{\partial P}{\partial x}(L, t)}. \quad (2.17)$$

For the same negative pressure gradient, $\partial P(L, t) / \partial t > 0$, as expected.

2.6. Continuous inventory and equilibrium pressure

The total ideal-gas mass is

$$m_{\text{tot}} = C_g \left[V_0 P_u + \varepsilon S \int_0^L P(x, t) dx + V_1 P_d \right]. \quad (2.18)$$

Integrating equation (2.6) over the core and adding equations (2.12) and (2.16) cancels both boundary fluxes and gives

$$\frac{dm_{\text{tot}}}{dt} = 0. \quad (2.19)$$

For the initial state in equation (2.10), the final uniform equilibrium pressure is consequently

$$\boxed{P_{\text{eq}} = \frac{V_0 P_{u,i} + (V_1 + \varepsilon SL) P_{d,i}}{V_0 + V_1 + \varepsilon SL}}. \quad (2.20)$$

This result is independent of k_ℓ , b , and μ : those parameters control the rate of equilibration, not the final ideal-gas mass balance. Equation (2.20) is therefore a useful numerical verification target.

3. Pressure transformation

3.1. Definition and derivative identities

Introduce

$$\boxed{\varphi = (P + b)^2}. \quad (3.1)$$

Because the model uses positive absolute pressure and $b \geq 0$,

$$P + b = \sqrt{\varphi}, \quad \boxed{P = \sqrt{\varphi} - b}. \quad (3.2)$$

Differentiation gives

$$\frac{\partial \varphi}{\partial x} = 2(P + b) \frac{\partial P}{\partial x}, \quad (3.3a)$$

$$\frac{\partial^2 \varphi}{\partial x^2} = 2 \left[\left(\frac{\partial P}{\partial x} \right)^2 + (P + b) \frac{\partial^2 P}{\partial x^2} \right], \quad (3.3b)$$

$$\frac{\partial \varphi}{\partial t} = 2(P + b) \frac{\partial P}{\partial t} = 2\sqrt{\varphi} \frac{\partial P}{\partial t}. \quad (3.3c)$$

In particular,

$$(P + b) \frac{\partial P}{\partial x} = \frac{1}{2} \frac{\partial \varphi}{\partial x}, \quad \frac{\partial}{\partial x} \left[(P + b) \frac{\partial P}{\partial x} \right] = \frac{1}{2} \frac{\partial^2 \varphi}{\partial x^2}. \quad (3.4)$$

This is the reason for the substitution: the nonlinear product in the spatial flux becomes one second derivative of φ .

3.2. Transformed core equation

Substituting equation (3.4) into the conservative pressure equation equation (2.8) gives

$$\varepsilon \frac{\partial P}{\partial t} = \frac{k_\ell}{2\mu} \frac{\partial^2 \varphi}{\partial x^2}. \quad (3.5)$$

From equation (3.3c),

$$\frac{\partial P}{\partial t} = \frac{1}{2\sqrt{\varphi}} \frac{\partial \varphi}{\partial t}. \quad (3.6)$$

Combining these two relations and cancelling 1/2 yields

$$\frac{\varepsilon}{\sqrt{\varphi}} \frac{\partial \varphi}{\partial t} = \frac{k_\ell}{\mu} \frac{\partial^2 \varphi}{\partial x^2}. \quad (3.7)$$

Thus,

$$\boxed{\frac{\partial \varphi}{\partial t} = \frac{k_\ell}{\varepsilon \mu} \sqrt{\varphi} \frac{\partial^2 \varphi}{\partial x^2}}, \quad 0 < x < L, \quad t > 0. \quad (3.8)$$

The equivalent form used by Jannot et al. (2007) is

$$\frac{\partial^2 \varphi}{\partial x^2} = \frac{\alpha}{\sqrt{\varphi}} \frac{\partial \varphi}{\partial t}, \quad \alpha \equiv \frac{\varepsilon \mu}{k_\ell}. \quad (3.9)$$

3.3. Transformed initial and boundary conditions

The initial conditions become

$$\varphi(0, 0) = (P_{u,i} + b)^2, \quad (3.10a)$$

$$\varphi(x, 0) = (P_{d,i} + b)^2, \quad 0 < x \leq L. \quad (3.10b)$$

For the upstream tank, use $(P + b)P_x = \varphi_x/2$ in equation (2.14), then multiply by $2\sqrt{\varphi}$:

$$\boxed{\frac{\partial \varphi}{\partial t}(0, t) = \frac{k_\ell S}{\mu V_0} \sqrt{\varphi(0, t)} \frac{\partial \varphi}{\partial x}(0, t)}. \quad (3.11)$$

The downstream condition is

$$\boxed{\frac{\partial \varphi}{\partial t}(L, t) = -\frac{k_\ell S}{\mu V_1} \sqrt{\varphi(L, t)} \frac{\partial \varphi}{\partial x}(L, t)}. \quad (3.12)$$

3.4. What the substitution does and does not do

The transformation removes the product $(P + b)P_x$ from the spatial operator and therefore makes a centred second-difference stencil natural. It does *not* convert the model into constant-coefficient linear diffusion, because

$$D(\varphi) = \frac{k_\ell}{\varepsilon \mu} \sqrt{\varphi} = \frac{k_\ell}{\varepsilon \mu} (P + b) \quad (3.13)$$

still depends on the solution.

Nor is φ proportional to stored gas mass. Under the ideal-gas assumption, storage is proportional to $P = \sqrt{\varphi} - b$, not to φ . Therefore, a pointwise finite-difference update of φ does not by itself guarantee exact discrete mass conservation. This distinction will be central when the finite-volume method is derived.

4. Explicit finite-difference formulation

4.1. Grid and discrete notation

Divide the core into N equal intervals:

$$x_i = i\Delta x, \quad i = 0, 1, \dots, N, \quad \Delta x = \frac{L}{N}. \quad (4.1)$$

Use uniform time levels

$$t^n = n\Delta t, \quad n = 0, 1, 2, \dots, \quad (4.2)$$

and the shorthand

$$\varphi_i^n \equiv \varphi(x_i, t^n). \quad (4.3)$$

The endpoint unknowns are shared with the tanks:

$$\varphi_0^n = \varphi_u^n, \quad \varphi_N^n = \varphi_d^n. \quad (4.4)$$

4.2. Interior nodes

At an interior node $i = 1, \dots, N - 1$, approximate the time derivative by forward Euler,

$$\left. \frac{\partial \varphi}{\partial t} \right|_i^n = \frac{\varphi_i^{n+1} - \varphi_i^n}{\Delta t} + \mathcal{O}(\Delta t), \quad (4.5)$$

and the second spatial derivative by the centred difference

$$\left. \frac{\partial^2 \varphi}{\partial x^2} \right|_i^n = \frac{\varphi_{i+1}^n - 2\varphi_i^n + \varphi_{i-1}^n}{\Delta x^2} + \mathcal{O}(\Delta x^2). \quad (4.6)$$

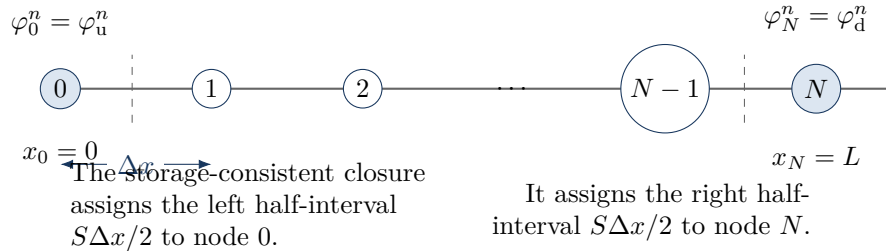


Figure 4.1.: Node-centred finite-difference grid. The shaded endpoint nodes are simultaneously the tank pressures and the core-face pressures.

The nonlinear coefficient $\sqrt{\varphi}$ is evaluated at the old time level. Substitution into equation (3.8) gives

$$\frac{\varphi_i^{n+1} - \varphi_i^n}{\Delta t} = \frac{k_\ell}{\varepsilon \mu} \sqrt{\varphi_i^n} \frac{\varphi_{i+1}^n - 2\varphi_i^n + \varphi_{i-1}^n}{\Delta x^2}. \quad (4.7)$$

Define

$$r_i^n = \frac{k_\ell \sqrt{\varphi_i^n} \Delta t}{\varepsilon \mu \Delta x^2} = \frac{k_\ell (P_i^n + b) \Delta t}{\varepsilon \mu \Delta x^2}. \quad (4.8)$$

Solving for the new value produces the explicit update

$$\varphi_i^{n+1} = \varphi_i^n + r_i^n (\varphi_{i+1}^n - 2\varphi_i^n + \varphi_{i-1}^n). \quad (4.9)$$

Equivalently,

$$\varphi_i^{n+1} = r_i^n \varphi_{i-1}^n + (1 - 2r_i^n) \varphi_i^n + r_i^n \varphi_{i+1}^n. \quad (4.10)$$

4.3. Direct dynamic-boundary update

This is the boundary discretization derived directly from equations (3.11) and (3.12).

4.3.1. Upstream

Use the one-sided derivative

$$\frac{\partial \varphi}{\partial x}(0, t^n) \approx \frac{\varphi_1^n - \varphi_0^n}{\Delta x}. \quad (4.11)$$

Forward Euler in equation (3.11) then gives

$$\varphi_0^{n+1} = \varphi_0^n + r_0^n (\varphi_1^n - \varphi_0^n), \quad (4.12)$$

where

$$r_0^n = \frac{k_\ell S \sqrt{\varphi_0^n} \Delta t}{\mu V_0 \Delta x} = \frac{k_\ell S (P_0^n + b) \Delta t}{\mu V_0 \Delta x}. \quad (4.13)$$

In weighted form,

$$\varphi_0^{n+1} = (1 - r_0^n) \varphi_0^n + r_0^n \varphi_1^n. \quad (4.14)$$

4.3.2. Downstream

At $x = L$, use

$$\frac{\partial \varphi}{\partial x}(L, t^n) \approx \frac{\varphi_N^n - \varphi_{N-1}^n}{\Delta x}. \quad (4.15)$$

The negative sign in equation (3.12) gives

$$\varphi_N^{n+1} = \varphi_N^n + r_N^n (\varphi_{N-1}^n - \varphi_N^n), \quad (4.16)$$

where

$$r_N^n = \frac{k_\ell S \sqrt{\varphi_N^n} \Delta t}{\mu V_1 \Delta x} = \frac{k_\ell S (P_N^n + b) \Delta t}{\mu V_1 \Delta x}. \quad (4.17)$$

Thus,

$$\varphi_N^{n+1} = r_N^n \varphi_{N-1}^n + (1 - r_N^n) \varphi_N^n. \quad (4.18)$$

4.4. Storage-consistent boundary update

The direct boundary formulas treat the endpoint node as tank storage only. On a node-centred grid, however, the core also contains a left and a right half-interval. Assigning those pore volumes to the corresponding endpoint nodes gives the effective storage volumes

$$V_{0,\text{eff}} = V_0 + \frac{\varepsilon S \Delta x}{2}, \quad (4.19a)$$

$$V_{1,\text{eff}} = V_1 + \frac{\varepsilon S \Delta x}{2}. \quad (4.19b)$$

Repeating the boundary update with these capacities gives

$$\varphi_0^{n+1} = \varphi_0^n + \hat{r}_0^n (\varphi_1^n - \varphi_0^n), \quad (4.20a)$$

$$\varphi_N^{n+1} = \varphi_N^n + \hat{r}_N^n (\varphi_{N-1}^n - \varphi_N^n), \quad (4.20b)$$

with

$$\hat{r}_0^n = \frac{k_\ell S \sqrt{\varphi_0^n} \Delta t}{\mu \Delta x (V_0 + \varepsilon S \Delta x / 2)}, \quad (4.21a)$$

$$\hat{r}_N^n = \frac{k_\ell S \sqrt{\varphi_N^n} \Delta t}{\mu \Delta x (V_1 + \varepsilon S \Delta x / 2)}. \quad (4.21b)$$

To show the exact connection with Jannot et al. (2007), define

$$\alpha = \frac{\varepsilon \mu}{k_\ell}, \quad \beta = \frac{\mu V_0}{k_\ell S}, \quad \delta = \frac{\mu V_1}{k_\ell S}. \quad (4.22)$$

Then

$$\hat{r}_0^n = \frac{\Delta t \sqrt{\varphi_0^n}}{\Delta x (\beta + \alpha \Delta x / 2)}, \quad \hat{r}_N^n = \frac{\Delta t \sqrt{\varphi_N^n}}{\Delta x (\delta + \alpha \Delta x / 2)}, \quad (4.23)$$

which reproduces their published endpoint equations after the indices are shifted from $1, \dots, m$ to $0, \dots, N$.

Interpretation. Both boundary versions approach the same continuum boundary condition as $\Delta x \rightarrow 0$. At finite resolution they are not identical:

$$\hat{r}_0^n = r_0^n \frac{V_0}{V_0 + \varepsilon S \Delta x / 2}, \quad \hat{r}_N^n = r_N^n \frac{V_1}{V_1 + \varepsilon S \Delta x / 2}. \quad (4.24)$$

The storage-consistent coefficients are therefore smaller. The difference is negligible only when each tank volume is much larger than the pore volume of one boundary half-interval.

4.5. Initialization, simultaneous update, and pressure recovery

The discrete initial state is

$$\varphi_0^0 = (P_{u,i} + b)^2, \quad \varphi_i^0 = (P_{d,i} + b)^2, \quad i = 1, \dots, N. \quad (4.25)$$

Every right-hand side in a single explicit step must use the same old-time array $\{\varphi_i^n\}$. In-place left-to-right overwriting would create a different, partly implicit scheme. After all nodes have been evaluated into a new array, pressure is recovered from

$$\boxed{P_i^{n+1} = \sqrt{\varphi_i^{n+1}} - b}. \quad (4.26)$$

5. Stability and monotonicity

5.1. Why a local analysis is needed

The interior coefficient

$$r_i^n = \frac{k_\ell(P_i^n + b)\Delta t}{\varepsilon\mu\Delta x^2} \quad (5.1)$$

changes with both position and time. A classical von Neumann analysis applies directly only to a linear constant-coefficient scheme. The standard practical step is therefore to *freeze* the coefficient locally and ask how one Fourier component of a small numerical error evolves for a fixed value r .

5.2. Interior Fourier analysis

For frozen r , the error obeys the same linear update as the solution:

$$e_i^{n+1} = e_i^n + r(e_{i+1}^n - 2e_i^n + e_{i-1}^n). \quad (5.2)$$

Assume a single discrete Fourier mode

$$\boxed{e_i^n = G^n e^{i\theta i}}, \quad (5.3)$$

where i is the spatial index, $i = \sqrt{-1}$, θ is the dimensionless wave angle, and G is the one-step amplification factor.

The shifted values are

$$e_i^{n+1} = G^{n+1} e^{i\theta i} = G \left(G^n e^{i\theta i} \right), \quad (5.4a)$$

$$e_{i+1}^n = G^n e^{i\theta(i+1)} = \left(G^n e^{i\theta i} \right) e^{i\theta}, \quad (5.4b)$$

$$e_{i-1}^n = G^n e^{i\theta(i-1)} = \left(G^n e^{i\theta i} \right) e^{-i\theta}. \quad (5.4c)$$

Substitution into [equation \(5.2\)](#) gives

$$G \left(G^n e^{i\theta i} \right) = G^n e^{i\theta i} \left[1 + r \left(e^{i\theta} - 2 + e^{-i\theta} \right) \right]. \quad (5.5)$$

Cancelling the nonzero common mode $G^n e^{i\theta i}$ leaves

$$\begin{aligned} G &= 1 + r \left(e^{i\theta} - 2 + e^{-i\theta} \right) \\ &= 1 + 2r(\cos \theta - 1) \\ &= \boxed{1 - 4r \sin^2 \left(\frac{\theta}{2} \right)}. \end{aligned} \quad (5.6)$$

Stability of every frozen Fourier mode requires

$$|G| \leq 1. \quad (5.7)$$

Since $0 \leq \sin^2(\theta/2) \leq 1$ and $r \geq 0$, the most restrictive mode has $\sin^2(\theta/2) = 1$. Hence

$$-1 \leq 1 - 4r \leq 1 \implies \boxed{0 \leq r \leq \frac{1}{2}}. \quad (5.8)$$

5.3. Weighted-average interpretation

The same limit follows directly from [equation \(4.10\)](#). Its three weights sum to one:

$$r_i^n + (1 - 2r_i^n) + r_i^n = 1. \quad (5.9)$$

They are all nonnegative when

$$0 \leq r_i^n \leq \frac{1}{2}. \quad (5.10)$$

The new value is then a convex combination of the old node and its two neighbours. It cannot create a new local maximum or minimum in one step. If $r_i^n > 1/2$, the centre weight becomes negative; the coefficients still sum to one, but the update can overshoot and oscillate.

5.4. Interior time-step limit

Solving [equation \(5.10\)](#) for Δt gives the local limit

$$\Delta t \leq \frac{\varepsilon \mu \Delta x^2}{2k_\ell(P_i^n + b)}. \quad (5.11)$$

Define an upper pressure bound $P_{\max}$. For a monotone pulse decay, $P_{\max} = P_{u,i}$ is valid because no pressure should exceed its initial upstream value. A safe global interior restriction is then

$$\boxed{\Delta t_{\text{int}} = \frac{\varepsilon \mu \Delta x^2}{2k_\ell(P_{\max} + b)}}. \quad (5.12)$$

5.5. Direct boundary-node restrictions

The direct upstream update is

$$\varphi_0^{n+1} = (1 - r_0^n)\varphi_0^n + r_0^n\varphi_1^n. \quad (5.13)$$

It remains between φ_0^n and φ_1^n if

$$0 \leq r_0^n \leq 1. \quad (5.14)$$

This gives

$$\boxed{\Delta t \leq \Delta t_{\text{up}} \equiv \frac{\mu V_0 \Delta x}{k_\ell S(P_0^n + b)}}. \quad (5.15)$$

A pressure-independent bound replaces P_0^n by $P_{\max}$.

Similarly,

$$\Delta t \leq \Delta t_{\text{down}} \equiv \frac{\mu V_1 \Delta x}{k_\ell S(P_N^n + b)}. \quad (5.16)$$

An interior node exchanges with two neighbours, so its retained weight is $1 - 2r_i^n$ and its limit is $1/2$. A direct tank node exchanges through one connection, so its retained weight is $1 - r_0^n$ or $1 - r_N^n$ and its limit is 1.

5.6. Storage-consistent boundary coefficients

For the storage-consistent closure, monotonic endpoint updates require

$$0 \leq \hat{r}_0^n \leq 1, \quad 0 \leq \hat{r}_N^n \leq 1. \quad (5.17)$$

The corresponding upstream limit is

$$\Delta t \leq \frac{\mu \Delta x (V_0 + \varepsilon S \Delta x / 2)}{k_\ell S(P_0^n + b)}, \quad (5.18)$$

with the analogous expression containing V_1 downstream.

The interior restriction [equation \(5.12\)](#) already implies these endpoint convexity limits. To see this, insert $\Delta t \leq \Delta t_{\text{int}}$ into [equation \(4.21a\)](#) and use $P_0^n \leq P_{\text{max}}$:

$$\begin{aligned} \hat{r}_0^n &\leq \frac{\varepsilon S \Delta x}{2(V_0 + \varepsilon S \Delta x / 2)} \\ &= \frac{\varepsilon S \Delta x}{2V_0 + \varepsilon S \Delta x} \leq 1. \end{aligned} \quad (5.19)$$

The same proof holds downstream. This explains why [Jannot et al. \(2007\)](#) state the classical interior diffusion limit without a separate tank-volume restriction for their published boundary formula.

5.7. Practical step selection

For the direct boundary update, use

$$\Delta t = \eta_s \min(\Delta t_{\text{int}}, \Delta t_{\text{up}}, \Delta t_{\text{down}}), \quad 0 < \eta_s < 1. \quad (5.20)$$

A reasonable initial safety factor is $\eta_s = 0.8$.

For the storage-consistent version,

$$\Delta t = \eta_s \Delta t_{\text{int}} \quad (5.21)$$

is sufficient for the convexity arguments above, although all coefficients should still be calculated and checked at runtime.

5.8. Qualification of the result

The Fourier calculation is a frozen-coefficient analysis of the *interior* stencil. The boundary inequalities are sufficient no-overshoot conditions for the explicit endpoint updates. Together they provide a transparent and useful safe-step criterion, but they are not a complete spectral proof for the fully coupled nonlinear tank–core update matrix.

The conditions are also stronger than bare linear stability in one useful sense: they make every update a convex combination. This promotes positivity and prevents new extrema, properties that are particularly valuable for pressure diffusion.

Finally, because $\Delta t_{\text{int}} \propto \Delta x^2$, doubling the number of spatial intervals requires approximately four times as many explicit time steps over the same physical duration.

6. Implementation specification and verification

6.1. One-step algorithm

The first Python program should implement one time step exactly as follows:

1. Store the old transformed state $\varphi^n = (\varphi_0^n, \dots, \varphi_N^n)$ without modifying it.
2. Calculate every interior r_i^n from [equation \(4.8\)](#).
3. Calculate either the direct endpoint coefficients [equations \(4.13\) and \(4.17\)](#) or the storage-consistent coefficients [equation \(4.21\)](#).
4. Check the applicable inequalities:

$$\max_{1 \leq i \leq N-1} r_i^n \leq \frac{1}{2},$$

and either $r_0^n, r_N^n \leq 1$ or $\hat{r}_0^n, \hat{r}_N^n \leq 1$.

5. Evaluate every new value into a separate array φ^{n+1} .
6. Reject the step if any value is nonfinite or nonpositive.
7. Recover P_i^{n+1} using [equation \(4.26\)](#).
8. Only after all checks pass, replace the old state and advance time.

The initial code should expose the choice of boundary formula clearly, for example as

```
boundary_scheme = "storage_consistent"    or    "direct".
```

The storage-consistent version should be the default reference; retaining the direct version makes the effect of the endpoint capacity correction testable.

6.2. Discrete inventory diagnostic

The continuous conserved quantity in [equation \(2.18\)](#) is proportional to a pressure–volume inventory. On the node-centred grid, approximate it with the trapezoidal rule:

$$\mathcal{I}^n = V_0 P_0^n + V_1 P_N^n + \varepsilon S \Delta x \left[\frac{1}{2} P_0^n + \sum_{i=1}^{N-1} P_i^n + \frac{1}{2} P_N^n \right]. \quad (6.1)$$

Monitor the relative drift

$$E_{\mathcal{I}}^n = \frac{\mathcal{I}^n - \mathcal{I}^0}{\mathcal{I}^0}. \quad (6.2)$$

This is a diagnostic, not a theorem of exact conservation for the transformed FDM. Because storage is linear in P while the update advances $\varphi = (P + b)^2$, finite time and grid errors can produce

inventory drift. The drift should decrease under refinement. The later conservative FVM will instead use shared face fluxes and pressure-based storage so cancellation is built into its semidiscrete equations.

6.3. Required verification tests

Test	Setup	Required result
Uniform pressure	Set $P_{u,i} = P_{d,i}$.	Every node and both tanks remain unchanged to roundoff.
Zero permeability	Set $k_\ell = 0$; bypass the time-step formula because its limit becomes infinite.	No state value changes.
Correct initial direction	Use $P_{u,i} > P_{d,i}$ and execute one stable step.	P_0 decreases, P_N does not decrease, and the interior response propagates from upstream to downstream.
Bounds and positivity	Run with all convexity limits satisfied.	No pressure leaves the interval $[P_{d,i}, P_{u,i}]$; all $\varphi_i > 0$.
Equilibrium	Run until the pressure difference is negligible.	Every pressure approaches equation (2.20) ; residual mismatch decreases with refinement.
Inventory	Record equation (6.2) through time.	Drift remains small and tends toward zero as $\Delta t, \Delta x \rightarrow 0$.
Time refinement	Repeat with $\Delta t, \Delta t/2, \Delta t/4$ on the same grid.	Tank-pressure histories converge; the forward-Euler temporal error is first order in the asymptotic range.
Grid refinement	Repeat with $N = 10, 20, 40, 80$, reducing Δt consistently.	Tank histories, equilibrium error, and inventory drift converge.
Boundary comparison	Run the direct and storage-consistent endpoint formulas on each grid.	Their difference is visible on coarse grids and decreases as $\Delta x \rightarrow 0$.
Controlled instability	In a disposable test, choose Δt just above the interior limit.	Overshoot or oscillation demonstrates the role of the negative centre weight; this case must never be used for fitting data.

6.4. Outputs for the baseline solver

The minimal solver should retain:

1. $P_u(t) = P_0(t)$ and $P_d(t) = P_N(t)$;
2. the differential pressure

$$\Delta P(t) = P_u(t) - P_d(t); \quad (6.3)$$

3. selected pressure profiles $P(x, t)$, rather than every profile at every small explicit step;
4. the largest interior and boundary coefficients at each stored time;
5. the inventory error $E_{\mathcal{I}}(t)$.

Expected qualitative behaviour is:

$$P_u(t) \searrow P_{\text{eq}}, \quad P_d(t) \nearrow P_{\text{eq}}, \quad \Delta P(t) \rightarrow 0. \quad (6.4)$$

6.5. Accuracy statements to keep precise

The forward-Euler time approximation is first order. The centred interior second derivative is second order. The boundary treatment has a separate error structure and should not be described solely by quoting the interior stencil's order. Grid and time refinement, rather than a label alone, must demonstrate the observed accuracy of the coupled calculation.

Similarly, “stable,” “monotone,” and “mass conservative” are distinct claims:

- the frozen-coefficient Fourier criterion addresses interior linear stability;
- the convex-weight restrictions prevent one-step overshoot;
- inventory drift evaluates discrete mass conservation;
- comparison with experimental traces is validation of the physical model, not verification of the numerical code.

6.6. Deferred extensions

The baseline should be verified before adding:

- pressure- and temperature-dependent viscosity;
- real-gas $Z(P, T)$ and a cubic equation of state;
- measured, time-dependent tank temperatures;
- uncertain initial conditions and calibrated line/dead volumes;
- history matching for k_ℓ , b , and possibly porosity;
- experimental sampling times and pulse-selection logic;
- a cell-centred, shared-flux finite-volume solver.

The next theoretical stage is the finite-volume derivation from the integral mass balance. The FDM documented here will then serve as an independent reference implementation rather than being discarded.

Bibliography

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A. Dimensional checks

A.1. Core equation

The transformed variable has units

$$[\varphi] = \text{Pa}^2, \quad [\sqrt{\varphi}] = \text{Pa}. \quad (\text{A.1})$$

The nonlinear diffusivity in [equation \(3.13\)](#) has units

$$\begin{aligned} \left[\frac{k_\ell}{\varepsilon \mu} \sqrt{\varphi} \right] &= \frac{\text{m}^2 \text{Pa}}{\text{Pa s}} \\ &= \text{m}^2 \text{s}^{-1}. \end{aligned} \quad (\text{A.2})$$

Therefore,

$$\left[\frac{k_\ell}{\varepsilon \mu} \sqrt{\varphi} \varphi_{xx} \right] = \text{m}^2 \text{s}^{-1} \frac{\text{Pa}^2}{\text{m}^2} = \text{Pa}^2 \text{s}^{-1} = [\varphi_t]. \quad (\text{A.3})$$

A.2. Interior explicit coefficient

For [equation \(4.8\)](#),

$$[r_i^n] = \frac{\text{m}^2 \text{Pa s}}{\text{Pa s m}^2} = 1. \quad (\text{A.4})$$

Thus r_i^n is dimensionless, as required for a numerical weight.

A.3. Boundary explicit coefficient

For the direct upstream coefficient,

$$[r_0^n] = \frac{\text{m}^2 \text{m}^2 \text{Pa s}}{\text{Pa s m}^3 \text{m}} = 1. \quad (\text{A.5})$$

Replacing V_0 with $V_0 + \varepsilon S \Delta x / 2$ does not change the units of the storage-consistent coefficient.

A.4. Time-step limits

The interior limit has units

$$\left[\frac{\varepsilon \mu \Delta x^2}{k_\ell (P + b)} \right] = \frac{\text{Pa s m}^2}{\text{m}^2 \text{Pa}} = \text{s}. \quad (\text{A.6})$$

The direct upstream limit likewise gives

$$\left[\frac{\mu V_0 \Delta x}{k_\ell S (P + b)} \right] = \frac{\text{Pa s m}^3 \text{m}}{\text{m}^2 \text{m}^2 \text{Pa}} = \text{s}. \quad (\text{A.7})$$

Equilibrium inventory. Every numerator term in [equation \(2.20\)](#) has units Pa m^3 , whereas every denominator term has units m^3 . Thus P_{eq} has units of pressure, and multiplication of the pressure–volume inventory by C_g gives mass.